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FRESHMAN ENGINEERING DEPARTMENT

DESIGN THINKING INNOVATION AND ENTREPRENEURSHIP

IDEA EXPO

IDEAS | INNOVATE | IMPACT

Y26 BATCH

INAUGURATION CEREMONY



PRINCIPAL

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COURSE INSTRUCTORS

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Date : 16, Oct, 2026

Time : 11:00 am

Venue : Bachupally Campus

Campus : Bachupally



ECOQUEST

Gamified Environmental Education Platform

Learn. Play. Protect the Planet.

Design Thinking for Innovation & Entrepreneurship (DTIE)

Section : 13 | Project Group 4

Course Coordinator: Dr. G. Sudhakar

Team Members

Team Eco Quest — Core Team

- Member 1: Ritika — 2620090062
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- Member 4: Shloka — 2620030679
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Mentor: DTIE Faculty Mentor — Department of Multi Disciplinary Innovation & Entrepreneurship

Vision and Mission

Vision

- A generation of environmentally literate citizens who treat sustainability as a way of life — not a subject — powered by the universal language of play.

Mission

- Make environmental learning as engaging as gaming through quests, rewards, and real-world eco-challenges.
- Bridge the climate literacy gap by turning curriculum topics into bite-sized, curriculum-aligned game levels.
- Convert screen time into green time: every in-game action maps to a real-world eco-action (planting, saving water, waste segregation).
- Build a measurable impact loop for schools and parents — track knowledge gained AND behavior changed.

Problem Statement

The Problem

- Low knowledge, lower confidence: only 12% of teens feel they know "a lot" about the causes of climate change, and only 10% about solutions (EdWeek / Planet Media Youth Climate Literacy Snapshot, 2025).
- Classroom teaching alone is not enough: lecture-based environmental education builds exam memory, not eco-habits.
- Misconceptions persist: only 54% of teens correctly identify human-caused greenhouse gas emissions as the main driver of climate change.
- Engagement gap: students already spend hours daily on screens, but almost none of that screen time builds environmental awareness.

Core Insight

- If learning feels like a game, students choose it voluntarily. If the game rewards real-world green actions, play becomes planet-positive behavior.

Research / Facts / Market & Industry Study

Key Facts (Problem Validation)

- Only 12% of teens feel they know "a lot" about climate-change causes; just 42% recognize the scientific consensus (2025 Youth Climate Literacy Snapshot. n = 1,017 teens. US).
- 71% of teens learn about climate change from teachers — yet school programs lack engaging tools to convert lessons into action.
- A meta-analysis of school science education shows strong gains in climate knowledge and attitudes. but weak gains in action — exactly the gap gamification targets.

Market & Industry Study

- Game-based learning market: USD 16.2 Bn (2023) → USD 28.4 Bn (2026) → USD 64.5 Bn by 2030. CAGR ≈ 22% (Grand View Research).
- Gamification-in-education segment: USD 2.11 Bn (2026) → USD 25.33 Bn by 2034. CAGR ≈ 36.4% — one of the fastest-growing edtech niches (Market Data Forecast).
- India angle: NEP 2020 + smartphone-first students + school ESG/club mandates = large. under-served market for curriculum-aligned eco-learning games.

Target User - Customer

Primary Users (Learners)

- Students aged 10–16 (Grades 5–10): digital natives who love games but find environmental textbooks dull.

Customers (Who Pays)

- Schools & teachers: seek curriculum-aligned, measurable environmental education (NEP 2020 emphasis on experiential learning).
- Parents: want productive, values-building screen time for their children.
- NGOs & CSR programs: need engagement tools for sustainability campaigns; sponsor licenses for under-resourced schools.

User Persona — Quick Snapshot

- Aarav, 13: plays 1.5 hours of mobile games daily, has a "green club" at school he finds boring, would love a game where his scores plant real trees.

Competitors / Benchmarks

Competitor	What it does	Gap EcoQuest fills
Kahoot! / Quizizz	Quiz-based gamified classroom learning	Generic quizzes; no environmental focus or real-world eco-actions
Minecraft Education (Sustainability City)	Immersive build-and-learn worlds	High hardware/licensing cost; no habit-tracking or curriculum mapping for Indian schools
Forest / Ecosia	Focus timer plants trees; search engine plants trees	Raises awareness but teaches no environmental curriculum
Seek by iNaturalist	Species identification via camera	Great outdoors tool, but no game loop, levels, or school dashboards
EcoQuest (ours)	Curriculum-aligned eco-quests + rewards + verified real-world green actions	Combines learning, play, and measurable behavior change in one low-cost mobile app

Risks and Gap

Key Risks & Mitigation

- Content accuracy: mislead risk if eco-facts are wrong → vet content with environmental educators: align to NCERT/UNESCO frameworks.
- Screen-time concern from parents: app rewards offline eco-actions more than screen play: built-in healthy-usage limits.
- Fake "green action" claims: photo verification + teacher/parent confirmation for real-world quests.
- Free alternatives: differentiate via curriculum mapping, school dashboards, and verified impact reports.

Gaps in the Market (Our Opportunity)

- No popular Indian product combines game-based environmental learning with verified real-world action.
- Existing tools track knowledge, not behavior — Eco Quest tracks both.

Concept/Prototype/Product/Solution

Eco Quest — The Concept

- A mobile-first learning game where students complete curriculum-aligned environmental quests, earn "Eco Points", badges and leaderboard ranks — and unlock real-world impact.

How It Works

- Learn: 5–8 minute quest-levels on climate, water, waste, biodiversity and energy, mapped to school curriculum.
- Play: quizzes, puzzles, AR nature hunts, team missions and city-builder mechanics where the virtual world grows greener as you learn.
- Act: offline eco-challenges (segregate waste, save X liters of water, plant a sapling) verified by photo/teacher sign-off.
- Impact: points convert into real trees planted via NGO partners; students, schools and parents get impact dashboards.

Why it wins: students get fun and status, schools get measurable NEP-compliant outcomes, sponsors get verified CSR impact.

Images/Storyboard/Architecture

Storyboard — A Student's Journey

1. Sign up → choose avatar, join school class-code; home screen shows a grey, polluted virtual city.
2. First quest → a 5-min "Know Your Carbon Footprint" quiz; city regains colour as the student scores.
3. Real-world challenge → "Carry a cloth bag today" — snap a photo; teacher verifies; earn 50 Eco Points + badge.
4. Team mission → class competes to reduce canteen plastic for a week; leaderboard updates live.
5. Impact → points fund a real sapling; student receives a geo-tagged "your tree" certificate.

System Architecture

Flutter mobile app → game engine (levels, points, badges) + offline quest cache.
Backend: Firebase/Node API, content CMS, verification service, analytics dashboard (schools/parents/sponsors).

Lean Canvas Model

PROBLEM	SOLUTION	UVP	UNFAIR ADVANTAGE	CUSTOMER SEGMENTS
• Climate literacy gap • Boring eco-education • No link between learning & action	• Gamified eco-quests • Verified real-world challenges • Impact dashboards	"The only game where playing plants real trees" — learning + play + verified impact	Curriculum mapping, school networks, NGO verification partnerships	• Students 10–16 • Schools & teachers • Parents • CSR/NGO sponsors
KEY METRICS	None	None	None	CHANNELS
DAU/MAU, quests completed, verified eco-actions, trees planted, school retention	None None None			School partnerships, app stores, green-club networks, social media, CSR programs
COST STRUCTURE	None	None	None	REVENUE STREAMS
App dev & content, server costs, NGO verification, marketing	None None None			Freemium app, school licenses (₹/student/yr), sponsor-funded school seats

Capital (Financial)

Startup Capital Required (Phase 1: MVP, 6 months)

Item	Estimated Cost (INR)
App development (Flutter game MVP) & design	₹ 3,50,000
Curriculum & content creation (educators, reviewers)	₹ 1,00,000
Cloud hosting, tools & licenses	₹ 50,000
Pilot in 5 schools + marketing	₹ 1,50,000
Contingency (≈10%)	₹ 75,000
TOTAL	₹ 7,25,000

Funding: seed grant / college incubator + CSR sponsor. Revenue: freemium + school licenses (~₹150/student/year) + sponsored seats for under-resourced schools.

Roadmap / Timeline

Phase	Timeline	Key Milestones
Phase 1 — Discover & Define	Months 1–2	User interviews with students/teachers, curriculum mapping, paper prototype of quests
Phase 2 — MVP Build	Months 3–4	Flutter MVP: 20 quest levels, points/badges, 3 real-world verified challenges
Phase 3 — Pilot	Month 5	Pilot in 5 schools; measure learning gain, engagement (DAU), verified actions; iterate
Phase 4 — Launch & Grow	Month 6+	Public launch, school licensing, NGO tree-planting partnership, sponsor dashboards

12-month goal: 50 schools onboarded, 25,000 students, 1,00,000 verified eco-actions, 10,000 trees planted.

THANK YOU